HONE

A Proof-of-Compute Blockchain for Inference, Sensors, and Sovereign Identity

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This document describes HONE as it runs on mainnet on 2026-11-04 (around epoch 353,900). Every mechanism is labelled **live**, **gated** (built and in the node, waiting for its activation epoch) or **planned**. Section 12 lists them in one table. Where an earlier version of this paper and the code disagreed, the code won; the differences are recorded in `WHITEPAPER_IMPLEMENTATION_NOTES.md`.

Abstract

HONE is a blockchain where every token in existence was earned by a machine doing useful work. Mining on HONE does not mean burning electricity on a hash that disappears the moment it is found. It means keeping the clock alive that the network depends on, storing data that someone committed, answering an AI inference job that a user submitted, or reporting sensor information that the network did not already have. The work had a customer.

Total supply is fixed at **42,000,000 HONE** with 10 decimal places (**1 HONE = 10,000,000,000 hunits**). Epochs run every **30 seconds** and are sealed by a quorum of permissionless clock nodes. There is no pre-mine and no founder allocation.

The chain is built for ordinary hardware. A phone is a full participant: it is a clock seat, a sensor, a wallet and, in shadow mode today, a juror. A Raspberry Pi, a PC and a GPU server run the same node binary with different profiles. A Flipper Zero is a signed sensor that reaches the chain through a phone that hears it.

Three ideas carry the 2026 design:

1. **Identity is owned, layered and private.** An account is a name held as an NFT, controlled by role keys (owner, active, posting, memo, hide, seek). Devices get their own seats under the owner. Devices broadcast rotating IDs that only their owner can link.
2. **Sensors are paid for information, not volume.** A report earns only when it is new or shows a change, the first reporter is paid, confirmations are paid near zero, and a claim pays out only once it is proven. Repeats earn nothing. Data older than seven days earns nothing.
3. **Disputes go to a jury, not to an operator.** Small local models on staked devices cast sealed votes on every judgment entry, and a seed-drawn jury reveals them when a result is challenged.

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1. The Problem

Compute is rented. AI inference is concentrated in a handful of providers who set price, policy and rate limits. Every prompt passes through someone else's servers.

Work is wasted. Bitcoin's security model works, but its computation produces nothing beyond the hash. Networks that tried to pay for "useful" work usually paid for volume, and volume is easy to fake.

Sensor networks pay for noise. Earlier decentralized sensor and coverage networks paid per reading or per beacon. Operators learned to send more readings, from the same place, saying the same thing. The data buyer gets redundancy. The honest operator gets diluted. HONE's own first sensor reward (`sensor_score(reading_count)`) had the same flaw, and it is being replaced (Section 6.4).

Tracking is the default. Bluetooth and Wi-Fi devices broadcast stable identifiers. A network that pays people to log those identifiers builds a surveillance system by accident. HONE has to count the world's devices without being able to follow any of them.

Keys get lost. The first HONE genesis ("Mayday", 1 May 2026) let wallets be created without durable key storage. Seed phrases were shown once and lost, leaving accounts that were owned but unsignable. A chain that claims sovereignty has to make recovery a precondition, not an afterthought.

Mobile nodes are behind NAT. Phones roam between networks and sit behind carrier NAT. A mesh that only works on one LAN stalls as soon as its phones leave the house. This happened on 2026-11-04 and is the reason HONE now runs a public node and is adding NAT traversal (Section 8).

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2. Design Overview

2.1 The chain in one page

Property	Value	Status
Chain ID	`hone` (testnet `hone-testnet`)	live
Genesis	4 July 2026, noon Los Angeles (`1783191600000` ms, 19:00 UTC)	live
Epoch	30 s (era 0)	live
Supply cap	42,000,000 HONE, block reward 2 HONE per epoch	live
Consensus	quorum seal by clock nodes, at least 51% of registered clocks	live
Storage engine	RocksDB, one shared block cache	live
Transport	libp2p: TCP + Noise, QUIC, WebSocket; gossipsub, Kademlia, mDNS	live
Keys	ed25519, SLIP-10 `m/44'/6942'/role'/0'`	live
Licence	Usufruct License (UFL), see `LICENSE`	n/a

2.2 Node types

One binary, `hone-node`, runs everywhere. What differs is the profile and the roles the operator turns on.

- **Full node (PC, server):** clock, storage, sensor relay, API, optional inference front door. Default profile.
- **Light node (Pi-class, under 2 GiB RAM):** the light profile is selected automatically below 2 GiB, or with `HONE_PROFILE=light`. It shrinks the RocksDB cache and memtables, slows catch-up vote traffic, and defaults storage, worker and embedded inference off. It never changes what consensus applies. Pi-based hotspot hardware such as Nebra units run this profile as full clocks.
- **Public relay node:** a non-clock node on a public IP that holds no posting key and only relays gossip and serves the public wallet API (Section 8.3).

- **Phone:** the Android app runs an embedded node as a **device clock seat**, a full sensor, a wallet and a jury seat (Section 5.2).
- **Flipper Zero:** a signed sensor with no network of its own. Its frames reach the chain through a phone (Section 5.3).

2.3 Hardware tiers

Inference rewards are weighted by a hardware tier from 0 to 4 with weights **1, 4, 8, 16, 32**. Today the tier is declared by the operator (`HONE_HW_TIER`) and is not attested; attested tiers are planned. The capability provisioner (Section 2.5) gates each role against detected RAM, VRAM and disk before anything heavy is downloaded.

2.4 Inference runs beside the node, not inside it

The node is the front door for inference: API, routing, pricing, fees, verification, payments and rewards all stay in the node and in consensus. **The node never loads a model in its own process.** Models run in a separate sidecar process, `hone-node inference-serve`, with its own systemd unit and memory budget, reached over loopback (`INFERENCE_URL`, default `127.0.0.1:4250`). The sidecar speaks the Ollama-compatible API (`/api/chat`, `/api/tags`, `/v1/models`, `/health`), so an operator can point the node at Ollama, vLLM or the bundled candle GGUF sidecar instead. The sidecar holds no keys, opens no chain store and joins no network. If it crashes or runs out of memory, inference calls fail and then recover; the node keeps sealing. A light node never runs a model at all.

This replaces the earlier design, in which inference ran in-process so that a verifier could rerun the job byte for byte. Keeping the clock alive turned out to matter more than in-process determinism. Verification now rests on verifier assessment and, for disputes, the decide jury (Section 7).

HONE is **model-agnostic**. There is no default model and nothing is downloaded automatically. The operator chooses what the machine serves.

2.5 The capability provisioner

A machine does not silently become an inference, storage, sensor or render node. The node discovers what it could do, gates each capability against its hardware, and the operator enables roles explicitly. Suggesting is never enabling. An on-chain model registry (`ModelManifestPublish`, keyed by the content hash of the weights, with a usage-weighted analysis reward) is **planned**; it is specified in `MODEL_REGISTRY_PROTOCOL.md` but has no ledger entry yet.

2.6 Native protocols

Four protocols are designed into HONE. Three are native ledger entry types processed by every node today. Each is a separable business owned initially by the protocol founder (shindevlin) and transferable by on-chain key rotation plus transfer of its GitHub repository.

- **Freeport (live):** a peer-to-peer marketplace with chain-enforced escrow. `StoreUpdate`, `ProductCreate`, `OrderPlace` (locks escrow), `OrderFulfill` (digital goods encrypted from the seller's `seek` key to the buyer's `hide` key), `OrderDispute`, `EscrowRelease`. Default settlement fee 0.5%.
- **Verasens (live):** the sensor network. `SensorRegister`, `SensorKeyRegister`, `SensorReading`, `SensorDataCommit` (a batch hash signed by the device key), `GatewayHeartbeat`, `CoverageReport`, and from the usefulness activation epoch `SensorSummary` (Section 6.4).
- **LinkGit (live):** Git-compatible version control. Repositories, refs and access grants are ledger entries; objects live on HONE storage nodes. Private repositories distribute their key to the grantee's `hide` key. A mirror protocol pushes to GitHub at the same time.

- **Wiiv (planned):** a rendering marketplace for image, video, audio and 3D work with milestone escrow and a reviewer market. Subjective quality is never a consensus input. The job and state-machine types exist in the `wiiv` crate; the ledger entries and chain routes are not wired yet. Its reserved accounts (`wiiv`, `wiiv-escrow`) exist from genesis.

Detailed entry tables live in `NATIVE_PROTOCOLS.md`, `FREEPORT_PROTOCOL_WHITEPAPER.md`, `VERASENS_PROTOCOL_WHITEPAPER.md`, `LINKGIT_PROTOCOL_WHITEPAPER.md` and `WIIV_PROTOCOL.md`.

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3. Identity

3.1 Names are the primary identity

A HONE account is first a human-readable name (`@bullship`), in the ENS and Hive tradition. A name costs a **10 HONE registration stake**, locked while the account exists, which keeps squatting expensive. System accounts (`__treasury__`, `__recycle_fund__`) have names but no key-derived address.

Typed, checksummed bech32 addresses for machines and tooling (`hh1…` accounts, `hk1…` contracts, `hd1…` devices and so on, specified in `ADDRESS_SCHEME.md`) are **planned**. They are not yet implemented in the node; today entries name accounts, and devices by their `owner@pubkey` seat.

3.2 Names are NFTs (live)

Since epoch **333,333** names are transferable assets (`IdentityNft*` entries). The model is deliberately simple: a name is a domain, and the chain never arbitrates identity. Each name carries a protection level chosen by its holder:

Level	Transferable	Recovery
Open	yes, by the active key	none
Guarded	yes, by the active key	guardians may recover to a destination pinned in advance
Locked	no	only guardian recovery to the pinned destination

The active key may only **tighten** protection (add recovery, lock, drop the founder backstop). Loosening needs the recovery authority. Recovery can only ever move a name to the destination pinned beforehand, so a guardian cannot redirect a name to itself, and a sequence number stops replay. A permanent lock can be lifted only by a 2-of-3 founder multisig. Open minting ends after the first 1,000 registrations; after that new names go to auction.

Reserved-name migration (live, epoch 352,000). At the first seal at or after epoch 352,000, every reserved-name account still controlled only by the genesis issuer key became a `NameNft` held by the issuer and bound to it. **1,162 names** migrated, with identical counts and state roots on every node. From then on such an account is controlled by whoever holds the NFT, and its value roles (owner, active) never fall back to a posting key.

3.3 Role keys

From one BIP-39 mnemonic a wallet derives six ed25519 role keys on the SLIP-10 hardened path `m/44'/6942'/role'/0'` (6942 is HONE's coin index). This is the curve and derivation that Ledger, Trezor and Keystone already sign for other ed25519 chains, so a hardware-wallet app is a build, not a redesign.

Key	Signs or decrypts	Where it should live
`owner`	key rotation, recovery	offline
`active`	transfers, stakes, escrow, memory writes	wallet with biometric unlock
`posting`	ordinary entries: listings, repo ops, sensor registration	hot
`memo`	encrypted memos, selective disclosure (location sharing)	wallet
`hide`	receives private content; root of device ID seeds; signs Find-My grants	wallet host only
`seek`	delivers encrypted content (store fulfilment, repo push)	delivery daemon

Since epoch 320,200 a missing role key falls back only to the posting key, and a signer with no usable key is rejected. Nodes hold their own **clock key**, not the owner's posting key: since epoch 349,100 node-originated entries (heartbeats, sensor registration, inference job bids and completions) may be signed by the registered clock key.

The same seed also derives EVM, Bitcoin (native SegWit), Solana and TON wallets client-side. HONE never takes custody of external assets.

3.4 Recoverable keystores

Every wallet writes an encrypted keystore (`<account>.keystore.json`, Argon2id + AES-256-GCM) and cannot be created without one. Recovery has three layers: the keystore file and password, the recovery phrase (shown once behind a write-it-down confirmation), and an optional ciphertext backup on a relay that cannot decrypt it.

3.5 Devices are seats under an owner

One account owns many devices, and each device is staked separately.

- **Clock seats (live since 321,000).** A device seat is the node ID `owner@pubkey`, where `pubkey` is the device's ed25519 key in hex. The ID binds the key, so only that device can sign for the seat, and only the owner may stake on it. Clock rewards earned by the seat are credited to the owner.
- **Decide seats (live since 352,440).** A device seat may also hold a `decide` role stake, staked by the owner's active key, and sign its own jury entries (Section 7).

3.6 Find-My grants (gated)

A phone should be able to find its owner's devices without holding the owner's hide key. The owner publishes a standing **`FIND_GRANT`** naming the phone's hardware key and up to 64 devices, signed by the owner's **current hide key** with no fallback to any other role. `FIND_REVOKE` withdraws it, and `SEED_GENERATION` rolls a device to a new ID seed generation. A grant is valid only while the hide key that signed it is still the account's hide key, so rotating `hide` voids every older grant with no state rewrite.

The phone side checks Android Key Attestation (Google root, TEE or StrongBox, the app's signing certificate) before the grant is issued. The per-device seed key is then imported into the phone's Keystore as a non-exportable key. The hide and owner keys never reach the phone. The entries are built and tested, with a cross-implementation fixture shared with the Android granting tool. **`FIND_GRANT_EPOCH` is not set yet**, so the entries are rejected until an activation epoch is approved. Until then a phone can still import the hide key directly, by the owner's choice.

4. Consensus

4.1 Clocks and quorum seals

Epochs are 30 seconds. Clock nodes propose the seal of each epoch, and an epoch is sealed when at least **51% of registered clocks** agree (rounded up, and never fewer than two for rewards). A node that loses its peers still seals locally so its own chain advances, but it earns nothing until a real quorum forms again. There is no solo-reward bypass, not even during bootstrap: delayed but recoverable rewards are better than divergent payouts.

During the first 250,000 epochs clocks could register at zero stake and build their minimum stake from clock rewards (the bootstrap grace). The minimum clock stake defaults to 5 HONE.

4.2 Seal lag (live since 351,880)

The quorum seal of epoch E commits only pending entries dated **E − 2 or earlier**. Before this rule, an entry gossiped in the last seconds of an epoch could land at E on its origin and at E+1 elsewhere. Balances agreed again a seal later, but every checkpoint compared states that differed by those in-flight entries.

4.3 Reward finality

An epoch's rewards, recycling and decay are applied only once the epoch is **20 epochs** (about 10 minutes) behind the tip. That is long enough that a heavier competing seal can no longer rewrite fork choice for it.

4.4 Surviving restarts and partitions (live)

Three fixes from October and November 2026 make a node's state independent of when it was restarted:

- **Persisted pending pool.** The pending pool and the outbox survive a restart, so a node's own heartbeat in flight at shutdown is no longer lost.
- **Backfill of entries and finalize meta.** A restarted or partitioned node asks peers for the quorum seals it missed and replays them, including each epoch's `EpochFinalize` meta. Without that meta, the reward driver stalled at the first gap. A stalled reward driver now triggers a finalize backfill, and migrations such as the 352,000 name migration run inside the backfill at their own epoch.
- **One dial per peer and a gossip watchdog.** Gossipsub sends topic subscriptions once, on a peer's first connection. With several parallel connections per peer, closing the one that carried the subscriptions left the peer invisible to gossip. The node now dials each peer once, and a watchdog disconnects and redials a connected peer that shows no topics for two 20-second checks. Separately, an operator guard restarts a node whose clock has not sealed for 15 minutes, at most once an hour.

4.5 Checkpoints and the self-healer (live; deterministic cursor planned)

Every **120 epochs** (one hour), right after applying a quorum seal, each clock records a checkpoint: the reward driver's cursor R, the balance Merkle root, and a snapshot of the sorted balance rows. It signs `hone-checkpoint:<chain>:<node>:<E>:<R>:<root>` with its clock key and gossips it.

A root is **canonical** for (E, R) if at least two distinct operators attest it and it has strictly more operators than any other root. Device seats never count, and one operator running many nodes counts once. Weights are deliberately absent, so one large staker cannot decide alone. A node

whose own root differs from a canonical root halts, fetches the snapshot, verifies its root, backs up its balances, rebases and resumes (`heal` mode). In `report` mode, the default, it only reports. With no canonical root it holds and never guesses.

Checkpoints are only comparable at the same cursor. A change that makes every node checkpoint at the same reward cursor (E − 21) is merged but not yet deployed; until it is, some checkpoints end in "hold" rather than "agree".

4.6 Genesis

Genesis was 4 July 2026, America's 250th anniversary. It seeded the founding protocol and operator accounts with preserved keys and **no balances**. Every founding key was created under the recoverable-keystore flow. Every node must use the identical genesis timestamp; a mismatched node forks at once and finds no peers.

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5. Sensors and Data

5.1 Rotating seeded device IDs

HONE devices never broadcast a stable identifier. Each device has an ID seed derived from the owner's hide key:

```
id_seed  = HKDF-SHA256(hide_private, "hone-id-seed" || device_pubkey [|| be32(generation)])
window   = floor(unix_utc / 900)                    15-minute UTC windows
period   = UTC calendar month of the window
share_k  = HKDF(id_seed, "hone-share" || period)
sk_w     = HKDF(share_k, "hone-pkw" || window)      X25519 secret for this window
pk_w     = X25519(sk_w, 9)                           broadcast in the advert
lid      = SHA-256(pk_w)[0..16]                      lookup id
```

- **Synced rotation.** Every device rotates at the same instant, on the UTC quarter-hour. Phones stop their advertising sets and start new ones, so the BLE address rotates with the key. A Flipper derives its two advert addresses from the same window key.
- **Advert v2.** Manufacturer data `'h' 'K' 02 win8 frag tag pk_w`: one extended advert, or two legacy halves joined by `(win8, tag)` rather than by address.
- **No stored seed.** The wallet mnemonic is the backup. Any device holding the hide key (or a Find-My grant) recomputes every owned device's keys.

Other people's devices are counted with keyed hashes: `oid = HMAC-SHA256(K_day, mac)[0..8]`, where `K_day` is 32 random bytes per observer per UTC day, kept in RAM and never stored or sent. An oid supports counts, dwell time and returns within one day on one observer. Nothing links across days or across observers.

5.2 Owner-encrypted sightings and location sharing (live in the app)

When an observer hears a complete `pk_w`, it seals one record for the owner and stores it under the lid:

```
record = {window, time, rssi, cell, accuracy, source}
blob   = epk || nonce || AES-256-GCM(HKDF(X25519(esk, pk_w)), record, aad = lid)
```

The observer keeps no plaintext location of anyone else's device. To anyone without `sk_w` the lid is random and cannot be linked across windows. The owner recomputes the lids of the last seven days (96 per device per day) and opens the blobs.

Location sharing uses the memo key. A grant is the owner's `share_k` for the current and next month, sealed to the grantee's on-chain memo key and stored in the grantor's memory service

(`MemorySet`, active-signed, about 500 bytes). The key name does not reveal who shares with whom. Revoking is `MemoryDelete`. A grantee who already fetched a month's key keeps it until the month ends; we state that limit plainly rather than hide it.

5.3 The phone as a full sensor (live)

The Android app is a complete sensor:

- **Location:** fused location, every reading kept and placed in the finest H3 cell its accuracy supports (res 11, 29 m, down to res 5). A private Wi-Fi fingerprint can stand in for GPS indoors.
- **Wi-Fi:** per access point `(oid, rssi, channel, band)` on 2.4, 5 and, where supported, 6 GHz.
- **BLE crowd:** distinct advertisers per window, as oids, with broad hardware filters so counts are the same screen-on and screen-off.
- **HONE sightings:** window keys heard from other HONE devices.
- **Flipper relay:** every Flipper frame the phone hears is stamped with the phone's cells.
- **Sound (off by default):** Perch v2 (Apache 2.0) runs on the phone over 5-second frames held only in RAM. If Perch detects any human class (speech, singing, crowd), the whole frame is dropped and only a counter goes up. What leaves the phone is species with confidence, noise levels and listening time. A public test that the speech filter misses less than 1% of speech is part of the plan before mic data can earn.
- **Clock seat, wallet and updates:** the phone is a device clock seat, sends HONE and stakes or unstakes role stakes with the active key held in the Android Keystore behind a biometric prompt for every signature, and updates itself from `https://honemesh.net/app/`.

Measured battery cost is about 3.5 mA on average indoors with the screen off (roughly 1.7% per day on a 5,000 mAh phone).

Signed device status every 15 minutes (planned). Each device will post a signed status report every **15 minutes**, aligned with the ID windows, and on state changes at most once a minute. A report carries counts, ages, on/off states and short error strings, and at most the H3 res-7 cell (about 5 km) of the latest fix. It never carries keys, audio, location trails or other people's IDs. The app side is in development; the collection endpoint is not live yet.

5.4 Flipper Zero (live)

The HONE Flipper firmware turns a Flipper Zero, optionally with a Wi-Fi board, into a signed sensor:

- **Signed v2 frames.** Every frame has an ed25519 signature by the device key. New bodies: BLE crowd, sightings and Wi-Fi v2 (top APs as oids, AP count, locally administered APs, probe senders). Frames are batched into signed commit groups (sub-GHz, Wi-Fi, tags, IR, BLE crowd, sightings) and archived on the SD card.
- **BLE observer.** The firmware ships the full ST BLE stack so it can scan: 8 seconds every 60, with day-keyed oids. Raw addresses never leave the scanning lane.
- **Sealed seed provisioning.** At claim time the owner's phone seals the device's ID seed to the Flipper's X25519 key. The Flipper opens it with its hardware AES and keeps the seed in RAM only.
- **No GPS.** A Flipper has no location. Its sightings carry `(lid, window, rssi, hits)`, and location comes only from a phone that hears it. Frames without a phone-stamped cell fill the archive but earn nothing under usefulness scoring.

5.5 Birds

The app identifies birds by sound (Perch v2) and photo, and a birds indexer (`/api/birds/*` on honemesh.net) collects sightings and shared photos and serves everyone's sightings back. **Live:**

the indexer and app sync. **Planned:** reference call playback (it needs an upstream API key), per-bird voice fingerprints, and public publishing of the bird, Flipper and coverage data sets.

5.6 Fixed listening stations (planned)

Many homes already have IP security cameras with microphones. A **`hone-listener`** add-on, packaged for Docker, Raspberry Pi, Home Assistant and Frigate, turns the audio track of a user's **own** cameras into fixed bird and sound stations.

- **Same pipeline as the phone.** First the speech gate, then Perch, then one report per 15-minute window. No audio is stored or sent; only species, noise levels and listening time leave the add-on.
- **One seat per camera.** Each camera is a sensor seat `owner@<station key>` under the owner's account. It is paid under the usefulness rules with deferred payout (Section 6.4).
- **Quality tiers** are set by the camera's audio:
  - **A:** 32 kHz or more, wideband.
  - **B:** 16 to 24 kHz.
  - **C:** 8 kHz (G.711), low-band species only.

  Every report carries its tier, and per-species thresholds take the tier into account.
- **Enrolment probes the stream.** When a camera is enrolled, the add-on measures its codec, sample rate, bandwidth and noise floor.
- **Low trust until corroborated.** A new station's claims stay at low trust until independent owners corroborate them.
- **Owned cameras only.** The add-on connects only to cameras the user owns. Access to other people's cameras, or to unsecured cameras found on a network, is never supported.

A fixed station complements phones: it listens from the same place every day, so arrivals, departures and changes stand out against a stable baseline.

5.7 The coverage map: two layers

	Public layer	Private layer
Who sees it	everyone	the owner only
Cells	H3 res 9 (about 170 m), coarser when accuracy is poor	finer, down to res 11
Contents	AP count, signal distribution, channel congestion, BLE and probe crowd estimates, HONE device count, noise, species, mobile-AP count	own devices' sightings, own routers, Wi-Fi fingerprints, Flipper frames
Identities	none: no MAC, BSSID, SSID, oid, lid or rid	owner's own, encrypted
On chain	signed per-(cell, window) summaries and batch hashes	never

Location trails never reach the chain. Each public summary carries a location weight (the sum of $1/\text{accuracy}^2$), so the map combines readings by inverse variance: better readings tighten the estimate, and a poor reading still contributes, just less. A person's own routers are registered privately by proof of control and appear only in the private layer.

6. Rewards and Economics

6.1 Supply and eras

The supply cap is **42,000,000 HONE**. The block reward is a constant **2 HONE per epoch** during the new-supply phase. Instead of halving the reward, HONE doubles the epoch length every 4,200,000 epochs, which halves daily emission:

Era	Epoch length	New supply per day	Duration
0	30 s	5,760 HONE	about 4 years
1	about 1 min	2,880 HONE	about 8 years
2	about 2 min	1,440 HONE	about 16 years
3	about 4 min	720 HONE	about 32 years
4	about 8 min	360 HONE	about 64 years
5+	recycle era	0 new supply	indefinite

Era 0's 30 seconds is the only fixed duration. At each era boundary the next era's epoch length is recomputed so that HONE's new supply ends with Bitcoin's last coin:

```
D_N = (target_end_ms - now_ms) / (4,200,000 x (2^remaining_eras - 1))
```

The target is Bitcoin's projected last block (block 6,930,000 at 10 minutes per block, about 2140). A Bitcoin height oracle (`BtcHeightReport`), which would let nodes that also run Bitcoin refine the target, is **planned**; today the compile-time projection is used.

6.2 Pools and layers

Each epoch's reward is split by measured work, not fixed percentages. Each pool (clock, inference, storage, sensor, verifier, service, mempool, tracker, runtime) has a calibration target, and its share is its utilization against that target, normalized across active pools. A pool with no claimants does not accumulate; its share recycles. Four layers combine:

- **Layer D, infrastructure base:** always fires. A small per-clock reward (0.001 HONE per clock at era 0), a testnet operator reward, and a mandatory reserve.
- **Layer B, activity pools:** utilization over target, with targets drifting slowly toward long-run 50% utilization.
- **Layer A, long-term scalar:** a fast and a slow moving average of utilization set the reward ceiling between 70% and 100% of the nominal schedule.
- **Layer C, fee boost:** verified fee volume from the previous epoch can raise emission toward the Layer A ceiling, with circular same-epoch flows netted out.

Inference fees split 80% to the worker, 15% to verifiers and 5% to recycle (70/10/15/5 with human reviewers when disputed).

6.3 No burn: the recycle fund

Tokens are never burned. Fees, surplus pool shares, slash proceeds and rounding remainders flow to `__recycle_fund__`. From era 5 the fund becomes the permanent reward source: each epoch it pays out 0.001% of its balance across the same pools. Because the draw is proportional to the balance, the fund self-corrects, and the 42,000,000 cap is never breached.

6.4 Sensors: pay only for useful information (gated)

One rule for every metric: pay for what is new or what changed. A change is an access point appearing in or disappearing from a cell, a value (signal, congestion, crowd, noise, species

activity) leaving its noise band, the same device seen in a different cell, or a species arriving or leaving.

A `SensorSummary` carries one public `(cell, window)` row of `metric -> value`. Scoring is deterministic and runs at apply, in commit order, on every node:

- **Novel** is the first report for (cell, metric, day bucket), or a value outside the metric's noise band around the chain's current estimate. Noise bands are fixed per metric (for example ±6 dB for signal strength, ±30% for crowd counts, exact for HONE device counts). The first valid reporter earns the novelty points (100), and the estimate moves.
- **Repeats earn 0.** One summary per device per metric per window, and one per *owner* per (cell, window, metric): a phone and a Flipper of the same owner count once.
- **Verification is paid near 0.** A later matching report by a different owner within ±2 windows earns 1 point and raises the original reporter's trust. One owner's devices never verify each other.
- **Contradiction:** a different owner in the same cell and window, outside the band, earns 0 and lowers the original reporter's trust.

Deferred payout. A novel claim is not paid when it arrives. It pays once it is proven: by an independent owner's corroboration, or by surviving a challenge window of **2,880 epochs (about 24 hours)** without a contradiction.

Old data pays nothing. Data older than **7 days** at submission fills the map but carries no claim.

Activation. `USEFULNESS_SCORING_EPOCH` is currently `u64::MAX`: the scoring code is built and tested, but inactive. Proven-point payout has its own gate, `USEFULNESS_PAYOUT_EPOCH`, also unset. An earlier activation at 352,700 was withdrawn so that deferred payout could be built first. **Until activation, the original volume scoring (`sensor_score(reading_count)`) still pays** for `SensorDataCommit`. Both gates switch together on every node at an approved epoch.

Anti-gaming rules beyond the above are planned: claw-back on contradiction, a location check that requires a report's Wi-Fi and BLE fingerprint to match the cell's known access points (GPS is spoofable), voiding windows that "teleport" faster than plausible travel, and escrow plus independent corroboration for rare acoustic events.

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7. The Decide Jury (shadow mode)

Some questions cannot be answered by a deterministic rule: is this coverage report useful, is this sensor reading plausible, is this entry normal. HONE answers them with a jury of small models on staked devices. It is designed so a juror cannot see others' votes before voting, cannot pick its cases, and cannot win by voting with the crowd.

7.1 Always-on sealed votes

Every judgment entry committed at a quorum seal gets **K = 7** voters, drawn deterministically from the epoch's seal hash among staked `decide` nodes with capacity left. Each voter asks its local `hone-decide` service, which returns an option and a confidence. The voter seals its votes as leaves of a Merkle tree (each leaf salted, over entry hash, epoch, task, option and confidence) and publishes only the root, as one **`DecideCommit` per epoch**. Votes stay hidden until revealed.

7.2 Random jury on dispute

A `DecideChallenge` opens a jury of **5 jurors plus 3 backups**, drawn from the seed of the **first sealed epoch after the challenge**, so nobody can know the jury when deciding whether to challenge. Jurors post `DecideReveal` with a Merkle proof against the root they committed earlier,

within **20 epochs**. A reveal that fails its proof is a strike. A **2/3 supermajority** of valid, non-abstaining reveals decides. Otherwise the case escalates to a larger model and then to the deterministic rule. The challenge window is 2,880 epochs.

At most one device per owner sits on any jury, across jurors and backups.

7.3 Spot checks and canaries

Three (node, entry) pairs per epoch are drawn as **spot checks** and must be revealed without any dispute, so a node that never really votes is caught. **Canaries**, items with a known answer, are shaped exactly like real items (unknown fields are rejected, so there is no canary flag to detect).

7.4 Models and confidence

The jury models are small students distilled from **Open-Jev** (MIT), at **135M and 360M** parameters, small enough to run on a phone. Each model ships a `thresholds.json` fitted on held-out data. A juror whose confidence is below the threshold for 95% accuracy (t95) **abstains** rather than guess. The pinned model is `smollm2-135m-r3` Q4, model hash `04df1989…2ab2ed`. Hashing the whole model directory, thresholds included, keeps every juror on the same model.

7.5 Stake and seats

A voting device holds a `decide NodeRoleStake`. The intended stake is **100 HONE per voting device**; the chain currently accepts any non-zero stake, and enforcing the minimum is planned. Device seats may hold a decide stake from epoch 352,440, and phones have light-seat API routes for work, commit and reveal.

7.6 Status

Decide entries are accepted from epoch **352,400**, in **shadow mode only**. They write `decide_*` meta records that are outside the state root, carry no fee, and change no balance, stake or other entry's validity. Outcomes are logged next to the deterministic rule, which is what actually holds. `DECIDE_JURY_BINDING_EPOCH` is unset. Jury outcomes become binding one task at a time, only after shadow results justify it. The usefulness tasks (`coverage_usefulness`, `sensor_plausibility_v2`) run beside usefulness scoring the same way.

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8. Networking

8.1 Transport

HONE uses libp2p: TCP with Noise and Yamux, QUIC for low-latency and mobile-friendly connections, and WebSocket for nodes behind restrictive networks. Gossipsub carries entries and seals, Kademlia finds peers, Identify and Ping keep connections honest, and mDNS finds peers on the local network. Peers identify themselves with a signed `hone-id` message (`GET /api/net/peers`).

8.2 Every node finds every other node

The requirement is simple: a phone, a PC and a Pi must find each other anywhere in the world, not only on one LAN. On 2026-11-04 the network learned what happens otherwise. Both phones left the LAN, a third clock went offline, quorum fell to 2 of 5, and finality stalled until a public path existed.

Discovery, in order:

1. **Peer cache** in RocksDB, peers seen in earlier sessions.

2. **Public-first bootstrap (live in the app and node configs).** Every node dials the public node at `/dns4/honemesh.net/tcp/6942` before anything else.
3. **DNS seed (planned).** `/dnsaddr/bootstrap.honemesh.net`, read from `_dnsaddr.bootstrap.honemesh.net` TXT records, so seeds change in DNS with no release. This replaces the earlier `bootstrap1` and `bootstrap2` names and the `_hone._udp` TXT design, none of which were ever published.
4. **Kademlia** walks the DHT from any connected peer.
5. **Censorship-resistant fallbacks:** a peer list on the Hive blockchain (live code), with TON and Bitcoin Ordinals registries designed and stubbed.

NAT traversal (planned, in development): circuit relay v2 (the public node as relay), AutoNAT to learn whether a node is reachable, DCUtR hole punching to upgrade relayed connections to direct ones, and UPnP port mapping on home routers.

8.3 The public node and wallet API (live)

A public, non-clock node runs at **honemesh.net**. It holds no posting key, relays gossip and serves a **public wallet API at `https://honemesh.net/chain/*`**. The API is an allowlist: reads (balances, accounts and history, role stakes, name resolution, entry status, memory, sensors, Find-My grants, supply, node info) and signed-entry submits (transfer, role stake and unstake, sensor commit and key registration, memory writes). Every other path returns 403. Bodies are capped at 256 KB and requests are rate-limited per IP. Clients sign locally; the public node never holds a user key.

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9. Privacy

HONE's privacy rules are design constraints, not settings.

- **No stable identifiers on air.** HONE devices rotate their ID and BLE address every 15 minutes on the UTC quarter-hour, all at once.
- **No linkable identifiers for anyone else.** Other devices are reduced to day-keyed hashes that cannot be linked across days or observers.
- **Sightings are for the owner.** Locations of a device are encrypted to that device's window key. Observers keep nothing they can read.
- **The chain sees aggregates.** Only per-(cell, window) summaries with no identities reach the chain. Location trails never do.
- **Sharing is explicit, scoped and expiring.** Location sharing is a memo-key grant per device per month, revocable, with its limit stated.
- **Audio never leaves the phone.** Frames live in RAM for five seconds. Any frame containing human sound is discarded whole. Mic sensing is off by default and needs the user's explicit acceptance of a notice. Whether using the mic is lawful where the user is, is the user's call.
- **Keys stay where they belong.** The hide and owner keys stay on the wallet host. A phone finds its owner's devices through a hardware-bound grant, and its active key is biometric-gated per signature.
- **Status reports are coarse.** The planned 15-minute device status carries at most a 5 km cell, never trails, audio, keys or other people's IDs.

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10. Security and Threat Model

Threat	Defence	Status
Unsigned spend for keyless accounts	role fallback only to posting; no-key signers rejected (320,200)	live
Compromised node steals funds	nodes hold a clock key; transfers and unstakes need the active key	live
Stolen name	NFT protection levels, pinned-destination recovery, founder backstop	live
One owner sybils quorum or checkpoints	checkpoints count distinct operators; device seats never count	live
Node drifts from consensus state	hourly signed checkpoints, self-healer halts and rebases	live (report mode default)
Restart or partition loses entries	persisted pending pool, entry and finalize backfill	live
Gossip silently stops	one dial per peer, gossip watchdog, stall guard	live
Phones isolated behind NAT	public node; relay, AutoNAT, DCUtR, UPnP	public node live; NAT traversal planned
Sensor volume farming	usefulness scoring; repeats 0; one reporter per owner	gated
Self-corroboration	one owner's devices never verify each other; one juror per owner	gated / shadow
Paid for unproven claims	deferred payout: corroboration or a 24-hour challenge window	gated
Stale data replay	no claim on data older than 7 days; duplicate and replay detection	gated
GPS spoofing, teleporting	fingerprint match to known APs; teleport voids window	planned
Lazy or colluding jurors	sealed commits, post-challenge jury seed, spot checks, canaries, abstain under t95	shadow
Inference crash takes down a node	models run in a separate sidecar with its own memory budget	live
Tracking through the sensor network	rotating IDs, day-keyed hashes, owner-encrypted sightings	live
Lost keys	mandatory recoverable keystore; mnemonic is the device-seed backup	live

Known limits, stated plainly. Hardware tiers are self-declared. Usefulness and jury outcomes do not move balances yet. A location-share grantee keeps a month key it already fetched. Until the deterministic checkpoint cursor is deployed, some checkpoints hold rather than agree. Until NAT traversal ships, phones off the LAN depend on the public node.

Operational rule. No balance is ever rewritten by hand without the founder's explicit approval.

One-time corrections use a dedicated, audited `state-correct` path.

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11. Governance and Activation Epochs

There is no on-chain governance token, council or multisig that can change HONE's fundamental parameters (epoch length, supply, coin type). Protocol parameters (fee rates, timeouts, storage pricing) are set by each protocol account's owner. Changes to consensus rules ship as **activation epochs**: a constant compiled into every node. Every node must run a binary that knows the rule before the epoch is reached, and the rule switches on at the same epoch everywhere. An activation epoch is set only after the founder approves it, and is kept well clear of checkpoints and at least an hour after the last node upgrades. An unapproved rule ships with its epoch set to `u64::MAX`, which means built, tested and off.

Constant	Epoch	Status
`CLOCK_BOOTSTRAP_GRACE_END_EPOCH`	250,000	past
`SIGNED_AUTH_ACTIVATION_EPOCH`	320,200	live
`DEVICE_CLOCK_ACTIVATION_EPOCH`	321,000	live
`IDENTITY_NFT_ACTIVATION_EPOCH`	333,333	live
`MEMORY_SIGNING_ACTIVATION_EPOCH`	349,100	live
`NODE_ENTRY_CLOCK_SIG_EPOCH`	349,100	live
`SEAL_LAG_ACTIVATION_EPOCH` (lag 2)	351,880	live
`NAME_NFT_MIGRATION_EPOCH`	352,000	live (1,162 names)
`DECIDE_ENTRIES_EPOCH`	352,400	live, shadow
`DECIDE_DEVICE_SEATS_EPOCH`	352,440	live
`USEFULNESS_SCORING_EPOCH`	unset (`u64::MAX`)	gated
`USEFULNESS_PAYOUT_EPOCH`	unset (`u64::MAX`)	gated
`FIND_GRANT_EPOCH`	unset (`u64::MAX`)	gated
`DECIDE_JURY_BINDING_EPOCH`	unset (`u64::MAX`)	gated

Future on-chain governance, if any, will be opt-in and weighted by contribution, not imposed.

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12. Implementation Status

As of 2026-11-04, branch `v2.1-live-identity`.

Area	Mechanism	Status
Consensus	30 s epochs, 51% quorum seals, no solo rewards	live
	Seal lag E−2	live (351,880)
	Reward finality depth 20	live
	Persisted pending pool and outbox	live
	Backfill of entries and finalize meta	live
	Hourly checkpoints, self-healer	live (report mode by default)
	Deterministic checkpoint cursor (E−21)	merged, not deployed

Area	Mechanism	Status
Identity	Six role keys, recoverable keystore	live
	Names as NFTs	live (333,333)
	Reserved-name migration, 1,162 names	live (352,000)
	Device clock seats `owner@pubkey`	live (321,000)
	Decide seats on devices	live (352,440)
	Find-My grants (FIND_GRANT, FIND_REVOKE, SEED_GENERATION)	gated, epoch not set
	Typed bech32 addresses (`hh1…`)	planned
Sensors	Rotating seeded IDs, 15-min synced rotation	live (app, Flipper)
	Keyed hashes for non-HONE devices	live
	Owner-encrypted sightings, memo-key location sharing	live (app)
	Phone full sensor (location, Wi-Fi, BLE crowd, sightings, Flipper relay)	live
	Perch v2 sound and bird sensing with speech dropped	live in app, off by default
	Birds indexer	live
	Flipper signed v2 frames, BLE observer, sealed seed, SD archive	live
	Signed device status every 15 minutes	planned (app side in development)
	Public bird, Flipper and coverage data sets	planned
	Fixed listening stations (`hone-listener` on owned IP cameras)	planned
	Two-layer coverage map	private layer live on device; public summaries queued for the summary commit
Rewards	Volume sensor scoring	live until usefulness activates
	Usefulness scoring, verification near 0, 7-day limit	gated
	Deferred payout (corroboration or 24 h window)	gated
	Claw-back, location fingerprint check, teleport voiding	planned
Jury	Sealed commits, Merkle root per epoch, post-challenge jury, spot checks, one device per owner	live, shadow

Area	Mechanism	Status
	135M / 360M Open-Jev-derived models with t95 abstain	live (pinned 135M)
	100 HONE minimum decide stake	intended; not enforced by the chain yet
	Binding outcomes	gated
Networking	TCP, QUIC, WebSocket, gossipsub, Kademlia, mDNS	live
	One dial per peer, gossip watchdog	live
	Public node and wallet API at honemesh.net	live
	Public-first bootstrap	live in configs and app
	DNS seed `bootstrap.honemesh.net`	planned
	Circuit relay, AutoNAT, DCUtR, UPnP	planned, in development
Inference	Separate sidecar process, node never loads models	live
	On-chain model registry	planned
Phones	Clock seat, sensors, wallet send and stake (biometric), in-app updater	live
Protocols	Freeport, Verasens, LinkGit	live
	Wiiv	planned (types only)
Economics	Era schedule, pools, recycle fund	live
	Bitcoin height oracle	planned

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13. Roadmap

Next (weeks):

1. Approve and activate `FIND_GRANT_EPOCH`, then switch phones to hardware grants.
2. Ship NAT traversal (relay, AutoNAT, DCUtR, UPnP) and publish the `bootstrap.honemesh.net` DNS seed, so phones on cellular networks keep quorum.
3. Deploy the deterministic checkpoint cursor and run the healer in heal mode everywhere.
4. Launch the 15-minute signed device status endpoint and its dashboard.
5. Set the usefulness scoring and payout epochs together, then resume the queued Flipper replay under the new scoring.

Then (months):

- Enforce the 100 HONE decide stake. Make the first jury task binding once shadow results justify it.
- Add claw-back, location fingerprint checks, teleport voiding and rare-event escrow.
- Publish the public bird, Flipper and coverage data sets, and bird voice fingerprints.
- Verifiable speech removal with a public test, before mic data earns.
- Fixed listening stations: the `hone-listener` add-on (Docker, Pi, Home Assistant, Frigate) for the user's own IP cameras, one sensor seat per camera, tiered A/B/C by sample rate.
- Attested hardware tiers. The on-chain model registry. Wiiv ledger entries.
- The Bitcoin height oracle.

Later:

- A HONE hardware-wallet app.
- Research pools where negative results also have value, and sensor data correlated with external events.

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14. Links

- **Website:** honemesh.net
- **White paper (PDF):** honemesh.net/HONE_WHITEPAPER.pdf
- **Android app:** honemesh.net/app/hone-latest.apk
- **Public wallet API:** `https://honemesh.net/chain/api/...`
- **GitHub:** github.com/shindevlin/hone
- **Native Protocols Overview:** NATIVE_PROTOCOLS.md
- **Sightings, coverage and rewards spec:** spec-sightings-coverage-rewards.md
- **Address Scheme:** ADDRESS_SCHEME.md
- **Chain constants:** CHAIN_CONSTANTS.md

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HONE v2.0, November 2026, Shin Devlin